The bounded elementary substructure relation is transitive

Silvia Barbina, Enrique Casanovas

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Editorial note. The counted statement is the author's environment at source0.tex lines 701-702 and its proof environment opens at line 703; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

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Editorial note. r002 note: r001 failed because the retained author proof cites two bibliography keys of the article; r002 bundles the article's own bibliography file and changes nothing else.

Editorial scope. One original result of the pinned author manuscript, counted once: transitivity of the bounded elementary substructure relation (author label sat3).

Allowed standard prerequisites (editorial).

- Bounded elementary substructure. A$\leq$n B means that A is elementary in B already after adding any set of at most n elements of B.. It is the relation whose transitivity is asserted.

- Diamond condition. For unconfined A1, A2 and B containing A1 with A2 elementary in B, the intersection of A1 and A2 is elementary in A1.. It is the tool that splits the added set.
- Closure of unconfined systems. The unconfined systems are closed under the relevant operations, so the intermediate systems of the argument remain unconfined.. It lets the diamond condition be applied.

Lemma 0.1 Let $A, A_1, A_2 \in \mathcal{C}_{\text{un}}$. Then

1. $\emptyset \leq A$
2. If $B \in \mathcal{C}_{\text{un}}$ is such that $A_1 \subseteq B$ and $A_2 \leq B$, then $A_1 \cap A_2 \leq A_1$.

Lemma 0.2 If $A \leq^n B$ and $B \leq^n C$, then $A \leq^n C$.

Proof. Assume $X \subseteq C \smallsetminus A$ and $|X| \leq n$. We check that $A \leq A \cup X$. Since $B \leq^n C$, we have $B \leq B \cup X$. By Lemma 0.1, $A \cup (X \cap B) \leq A \cup X$. Since $A \leq^n B$, we know that $A \leq A \cup (X \cap B)$. Hence $A \leq A \cup X$. □

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